Power BI Project Documentation

Cookie Sales and Customer Insight Dashboard

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1. Project Objective

The objective of this project is to design and develop an interactive business intelligence dashboard to analyze sales performance, profitability, and customer behavior. The dashboard enables stakeholders to make data-driven decisions by providing insights into revenue trends, cost structures, and customer purchasing patterns.

2. Project Overview

This project focuses on transforming raw transactional data into meaningful insights using data modeling, derived columns, and DAX measures. Two interactive dashboard pages were created:

- Sales & Revenue Performance Dashboard
- Customer Insights Dashboard

The solution helps in:

- Monitoring revenue and profit trends over time
- Evaluating product and category performance
- Understanding customer purchasing behavior
- Tracking key KPIs such as profit margin, average order value, and YTD revenue

3. Data Model & Relationships

The data model is structured using a star schema approach to ensure efficient querying and scalability.

Tables Used

- **Fact Table**: Sales/Orders Data
- **Dimension Tables**:
    - Date
    - Product (Category, Cookie Type)
    - Customer (Name, City)

Relationships

From Table	Column	To Table	Column	Relationship Type
Sales	Date	Date	Date	Many-to-One
Sales	Product ID / Category	Product	Category	Many-to-One
Sales	Product ID / Category	Customer	Name	Many-to-One

4. Derived Columns

Column Name	Logic Used	Purpose of Creation
Month	Extracted from Date	To analyze monthly sales trends
Year	Extracted from Date	To perform yearly comparisons
Profit	Revenue – Cost	To calculate profit at row level for detailed analysis

5. DAX Measures

Title	DAX Measure	Purpose of Creation
Total Revenue	`Total Revenue = SUM(Sales[Revenue])`	To calculate overall revenue generated
Total Cost	`Total Cost = SUM(Sales[Cost])`	To calculate total cost incurred

Total Profit	`Total Profit = SUM(Sales[Profit])`	To measure total profitability
Total Units	`Total Units = SUM(Sales[Units Sold])`	To track total quantity sold
Average Order Value	`AOV = DIVIDE([Total Revenue], COUNT(Sales[Order ID]))`	To measure average revenue per order
Profit Margin %	`Profit Margin % = DIVIDE([Total Profit], [Total Revenue]) * 100`	To evaluate profitability efficiency
Revenue per Customer	`Revenue per Customer = DIVIDE([Total Revenue], DISTINCTCOUNT(Sales[Customer Name]))`	To understand customer value
YTD Revenue	`YTD Revenue = TOTALYTD([Total Revenue], 'Date'[Date])`	To track cumulative revenue over the year

6. Python-Based Custom Visualization

Overview

A custom visualization was created using the Python script editor integrated within Power BI to enhance analytical capabilities beyond standard visuals. This chart combines both area and line visualization techniques to represent revenue and profit trends across different cookie types.

Purpose of Use

- To create a hybrid visualization (Area + Line chart) not readily available in default visuals
- To provide a clear comparison between revenue and profit in a single view

- To leverage Python's flexibility for advanced data transformation and visualization
- To improve storytelling by highlighting performance differences across products

Visualization Logic Explanation

- Data is grouped by Cookie Type to calculate total revenue and cost
- Profit is derived by subtracting cost from revenue
- An area chart is used to represent revenue distribution
- A line chart is overlaid to show profit trends
- Sorting ensures better readability by displaying highest revenue categories first

Python Script Used

```
import pandas as pd

import matplotlib.pyplot as plt

import matplotlib

matplotlib.use('Agg')


df = dataset


summary = df.groupby('Cookie Type').agg({

    'Revenue': 'sum',

    'Cost': 'sum'

}).reset_index()


summary['Profit'] = summary['Revenue'] - summary['Cost']

summary = summary.sort_values(by='Revenue', ascending=False)


plt.figure(figsize=(10,5))


# Area chart for Revenue

plt.fill_between(summary['Cookie Type'], summary['Revenue'],

                 color='violet', alpha=0.6, label='Revenue')
```

```
# Line chart for Profit
plt.plot(summary['Cookie Type'], summary['Profit'],
         color='red', marker='o', linewidth=2, label='Profit')

plt.xticks(rotation=45)
plt.title('Revenue and Profit by Cookie Type')
plt.legend()

plt.tight_layout()
plt.show()
```

7. Dashboard Overview

Page 1: Sales & Revenue Performance

KPI Cards:

- Total Revenue
- Total Cost
- Total Profit
- YTD Revenue

Visualizations:

1. **Revenue Trend**
    - Chart Type: Line Chart
    - X-Axis: Date
    - Y-Axis: Total Revenue
    - Purpose: To analyze revenue trends over time
2. **Profit Performance**
    - Chart Type: Waterfall Chart
    - Category: Product Category / Product
    - Y-Axis: Total Profit
    - Purpose: To visualize profit contribution across categories
3. **Revenue vs Profit**
    - Chart Type: Python Visual
    - Purpose: To compare revenue and profit using advanced visualization logic

Page 2: Customer Insights Dashboard

KPI Cards:

- Total Customers
- Total Units
- Profit Margin %
- Revenue per Customer
- Average Order Value

Visualizations:

1. **Orders from City**
    - Chart Type: Funnel Chart
    - Category: City
    - Values: Total Units
    - Purpose: To analyze order distribution across cities
2. **Customer Purchase Activity**
    - Chart Type: Donut Chart
    - Legend: Customer Name
    - Values: Total Revenue
    - Purpose: To identify top contributing customers
3. **Cookie & Revenue Analysis**
    - Chart Type: Bar Chart
    - Y-Axis: Cookie Type
    - X-Axis: Units Sold
    - Purpose: To analyze product-level performance
4. **Customer vs Product**
    - Chart Type: Bar Chart
    - X-Axis: Units Sold
    - Values: Cookie Type
    - Purpose: To understand product demand distribution

8. Key Insights Delivered

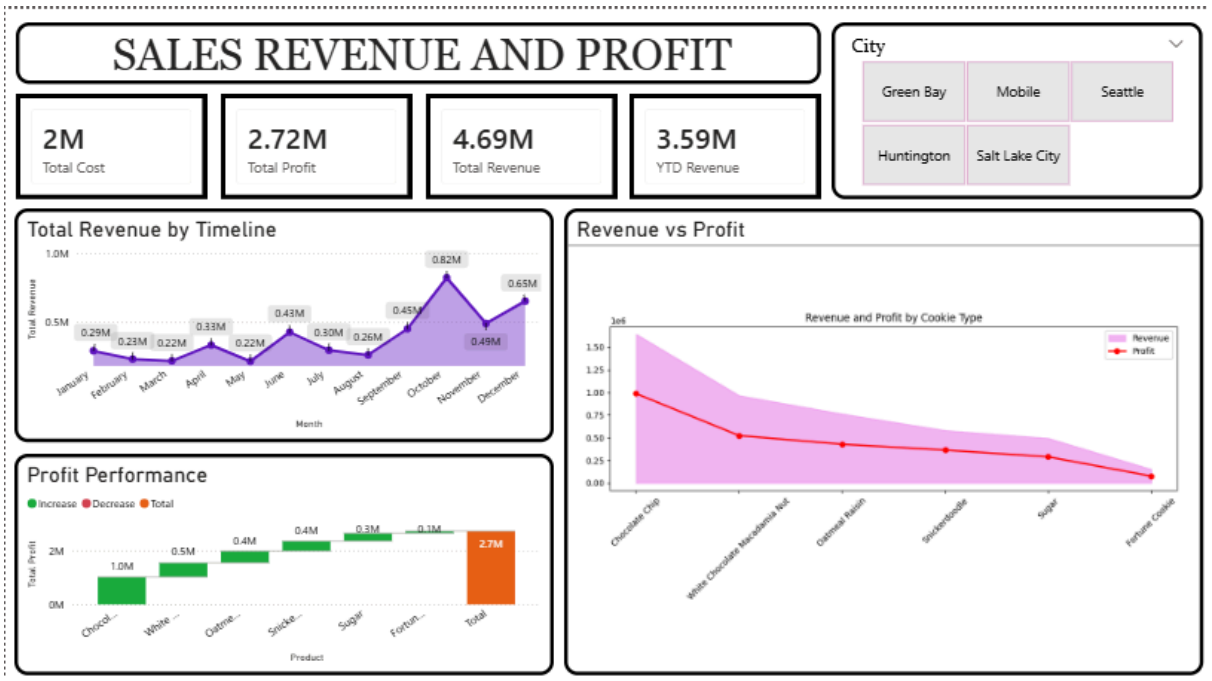
- Identified top-performing products and categories contributing the highest revenue and profit.
- Discovered seasonal sales trends using monthly and yearly analysis.
- Highlighted high-value customers driving a significant portion of revenue.
- Analyzed profitability efficiency through profit margin percentage.
- Evaluated city-wise demand distribution, helping in regional decision-making.

9. Tools & Technologies Used

Tool/Technology	Purpose
Power BI	Data visualization and dashboard creation
DAX (Data Analysis Expressions)	Creating measures and calculations
Power Query	Data transformation and cleaning
Python (Power BI Integration)	Advanced/custom visualizations
Excel/CSV Dataset	Data source for analysis

10. Conclusion

The developed dashboards provide a comprehensive view of business performance by combining sales, profitability, and customer analytics. The use of data modeling, calculated columns, and DAX measures ensures accurate insights and supports better decision-making.



CUSTOMER AND PRODUCTS INSIGHTS

Month

Apr | Jun
Aug | Mar
Dec | May
Feb | Nov
Jan | Oct
Jul | Sep

Product

Chocolate Chip
Fortune Cookie
Oatmeal Raisin
Snickerdoodle
Sugar

5
Total Customers

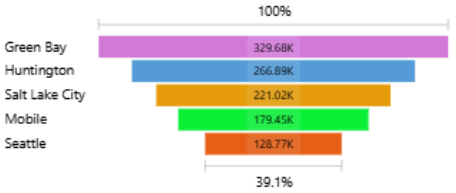
1.13M
Total Units

0.58
Profit Margin%

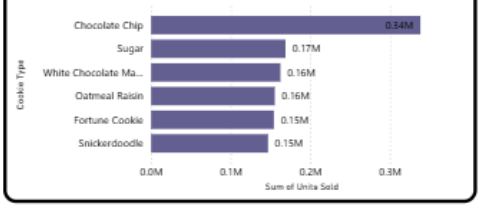
938.05K
Revenue Per Customer

6.70K
AVG order value

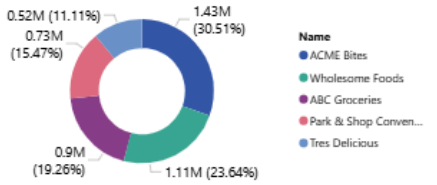
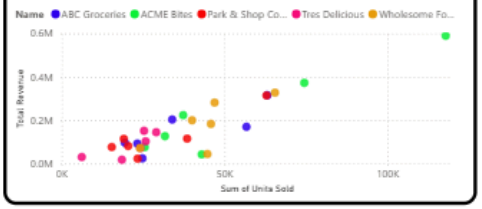
Orders from City



Cookie and Revenue



Customer Purchase Activity



Customer Vs product